

Practical 9 - Conditional logistic regression

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `vit_E.csv` and `infertility.csv`

Introduction

There are two parts to this session.

In the first part we re-visit the vitamin E dataset from the previous practical and apply the conditional logistic regression methods we have learnt about.

In the second part we use a new dataset and investigate the association between infertility in women and number of previous spontaneous and induced abortions.

Aims

- Learn how to fit a conditional logistic regression model to matched data.
- Learn how to fit a standard logistic regression model to matched data by including the matching variables.
- Understand the different assumptions underpinning the two models described above.

Part A. Vitamin E dataset

Reload the vitamin E dataset and generate a binary variable to indicate high vitamin E levels (above 12mg/dl).

As a reminder, in the previous session we found that the estimated conditional odds ratio for the association between the binary vitamin E variable and a person's cancer status (case or control) was 0.76, indicative of a protective effect of high vitamin E levels.

Question 1

We will first demonstrate an invalid method of analysis which might seem superficially attractive.

Fit a logistic regression model with case as the dependent variable, and the binary vitamin E indicator variable and the set variable as categorical explanatory variables. Confirm that this does not give the same (correct) estimate of the odds ratio as we calculated in the previous session (0.76).

This situation contrasts with the linear regression we used in the previous practical session, where adjusting for the matched set produced a valid estimate of the mean difference in vitamin E levels between cases and controls. This type of analysis is not valid when the dependent variable is binary.

SOLUTIONS

```
vit <- read.csv("vit_E.csv")
vit$vitEHigh <- 1*(vit$vitE>12)

setFactorMod <- glm(case ~ vitEHigh+factor(set),
                    family=binomial,
                    data=vit)
summary(setFactorMod)$coefficients[1:2,]
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.275412	1.4351126	0.1919097	0.84781297
vitEHigh	-0.550824	0.2929411	-1.8803230	0.06006407

```
exp(coef(setFactorMod)[2])
```

```
vitEHigh
0.5764746
```

This estimate of 0.58 is incorrect. It is actually equal to the square of the correct estimate ($0.7592593^2 = 0.5764746$), see the course manual for details. Adjusting for the matched set as a fixed effect in a logistic regression model is not a valid approach to the analysis of case-control studies.

Question 2

Fit a conditional logistic regression analysis to estimate the odds ratio for the effect of high vitamin C, and compare with the results from the previous practical.

SOLUTIONS

Following the code in the lecture slides, we can use:

```
library(survival)
clogit1 = clogit(case ~ vitEHigh + strata(set),
                 data = vit)
summary(clogit1)
```

Call:

```
coxph(formula = Surv(rep(1, 542L), case) ~ vitEHigh + strata(set),
      data = vit, method = "exact")
```

```
n= 542, number of events= 271
```

	coef	exp(coef)	se(coef)	z	Pr(> z)
vitEHigh	-0.2754	0.7593	0.2071	-1.33	0.184

	exp(coef)	exp(-coef)	lower .95	upper .95
vitEHigh	0.7593	1.317	0.5059	1.139

```
Concordance= 0.524 (se = 0.025 )
```

```
Likelihood ratio test= 1.78 on 1 df, p=0.2
```

```
Wald test = 1.77 on 1 df, p=0.2
```

```
Score (logrank) test = 1.78 on 1 df, p=0.2
```

Using conditional logistic regression recovers the correct estimate of the odds ratio. Note however that the 95% confidence interval is slightly different to the one obtained using the Mantel-Haenszel approach. The conditional logistic regression confidence interval relies on asymptotic results whereas the Mantel-Haenszel approach permits construction of an exact confidence interval.

Question 3

We could also use the original, continuous vitamin E variable as an explanatory variable. However, as the dependent variable is still binary, we must again use conditional logistic regression. Fit this model and give an interpretation for the results.

SOLUTIONS

```
clogit2 = clogit( case ~ vitE + strata(set),  
                  data = vit)  
summary(clogit2)
```

Call:

```
coxph(formula = Surv(rep(1, 542L), case) ~ vitE + strata(set),  
      data = vit, method = "exact")
```

n= 542, number of events= 271

	coef	exp(coef)	se(coef)	z	Pr(> z)
vitE	-0.01153	0.98853	0.02414	-0.478	0.633

	exp(coef)	exp(-coef)	lower .95	upper .95
vitE	0.9885	1.012	0.9429	1.036

Concordance= 0.546 (se = 0.043)

Likelihood ratio test= 0.23 on 1 df, p=0.6

Wald test = 0.23 on 1 df, p=0.6

Score (logrank) test = 0.23 on 1 df, p=0.6

Interpretation: Each 1mg/dl increase in vitamin E is associated with a 1.15% decrease ($0.0115 = 1 - 0.9885$) in the odds of cancer, conditional on age, duration of sample storage and smoking status (the matching variables). This association is not statistically significant ($p = 0.6$) with a 95% confidence interval that extends from a 5.71% decrease to a 3.64% increase per 1mg/dl increase in vitamin E.

Note that the p-values (likelihood ratio and Wald test) here are very similar to that from the paired t-test comparing vitamin E levels in cases and matched controls in the matched studies practical.

Question 4

Suppose now that we wish to explore whether the effect of dichotomised vitamin E on the conditional odds of cancer varies with observation time (time from blood collection to diagnosis of cancer in the case). Carry out an appropriate analysis to explore this. What do you conclude from this analysis?

SOLUTIONS

To investigate whether the odds ratio for **dichotomised** vitamin E differs according to the time between blood collection and diagnosis of cancer in the case, fit a model with an interaction between vitEHigh and time:

```
clogit3 = clogit( case ~ vitEHigh*factor(time) + strata(set),
                  data = vit)
summary(clogit3)
```

Call:

```
coxph(formula = Surv(rep(1, 542L), case) ~ vitEHigh * factor(time) +
      strata(set), data = vit, method = "exact")
```

n= 542, number of events= 271

	coef	exp(coef)	se(coef)	z	Pr(> z)
vitEHigh	-0.7340	0.4800	0.3512	-2.090	0.0366 *
factor(time)2	NA	NA	0.0000	NA	NA
factor(time)3	NA	NA	0.0000	NA	NA
vitEHigh:factor(time)2	0.4463	1.5625	0.5637	0.792	0.4285
vitEHigh:factor(time)3	0.8965	2.4510	0.4818	1.861	0.0628 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
vitEHigh	0.480	2.083	0.2412	0.9554
factor(time)2	NA	NA	NA	NA
factor(time)3	NA	NA	NA	NA
vitEHigh:factor(time)2	1.562	0.640	0.5176	4.7169
vitEHigh:factor(time)3	2.451	0.408	0.9533	6.3019

Concordance= 0.535 (se = 0.025)

Likelihood ratio test= 5.34 on 3 df, p=0.1

Wald test = 5.04 on 3 df, p=0.2

Score (logrank) test = 5.24 on 3 df, p=0.2

Note that the main effects of time cannot be estimated (R has dropped these) because this is a matching variable, but the interaction terms can be estimated.

If want to compute the estimated effect in the time=2 group and its SE:

```
time2logOR <- coef(clogit3)[1]+coef(clogit3)[4]
#use Var(A+B)=Var(A)+Var(B)+2*Cov(A,B)
#so SE is
time2logOR_SE <- sqrt(vcov(clogit3)[1,1]+vcov(clogit3)[4,4]+
  2*vcov(clogit3)[1,4])
exp(time2logOR)
```

```
vitEHigh
0.75
```

```
exp(time2logOR-1.96*time2logOR_SE)
```

```
vitEHigh
0.3160165
```

```
exp(time2logOR+1.96*time2logOR_SE)
```

```
vitEHigh
1.77997
```

To test whether there is evidence the vitamin E effect differs by time, we can compare the first of the last model with the one where the interaction was not included:

```
anova(clogit3, clogit1)
```

Analysis of Deviance Table

```
Cox model: response is Surv(rep(1, 542L), case)
Model 1: ~ vitEHigh * factor(time) + strata(set)
Model 2: ~ vitEHigh + strata(set)
    loglik   Chisq Df Pr(>|Chi|)
1 -185.17
2 -186.95 3.5555 2      0.169
```

Although the estimated odds ratio when the time between diagnosis and cancer diagnosis is shortest (time = 1) is statistically significant (p=0.037) the likelihood ratio test of the interaction is not statistically significant (p=0.169) so we should interpret this result cautiously.

We could instead use the continuous vitamin E variable:

```
clogit4 = clogit( case ~ vitE*factor(time) + strata(set),
                  data = vit)
anova(clogit4, clogit2)
```

Analysis of Deviance Table

```
Cox model: response is Surv(rep(1, 542L), case)
Model 1: ~ vitE * factor(time) + strata(set)
Model 2: ~ vitE + strata(set)
      loglik  Chisq Df Pr(>|Chi|)
1 -183.39
2 -187.73 8.6844  2    0.01301 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Here the evidence for an interaction, and for an association between vitamin E levels and cancer risk only when the time between blood collection and cancer diagnosis in the case is shortest, is stronger.

Part B. Infertility dataset

The data for Part B come from a matched case-control study of the association between infertility in women and previous spontaneous or induced abortion. This is an example of a matched case-control study with two controls per case. The dataset is called `infertility.csv`. The main research question of interest is whether previous abortions (induced or spontaneous) affect risk of infertility.

Further details of the study can be found in the original paper by Tzonou et al, which is available [online](#).

Each case (a woman diagnosed as infertile) was matched to two controls on age, parity and education level. The variables shown in the data are as follows:

Variable	Description
<code>case</code>	case/control indicator (0 = control, 1 = case)
<code>stratum</code>	index for matched set, numbered 1–271
<code>spontaneous</code>	Number of prior spontaneous abortions (0 (zero), 1 (one), 2 (2 or more))
<code>induced</code>	Number of prior induced abortions (0 (zero), 1 (one), 2 (2 or more))

Variable	Description
age	age in years
parity	number of children
education	years of education (1 (0–5 years), 2 (6–11 years), 3 (12+ years))

Question 5

Open the dataset and familiarise yourself with the data and its structure:

- Do rows contain information on individual people or matched sets?
- How many cases, controls, and matched sets are included in the dataset?
- Are all of the sets the same size?

SOLUTIONS

```
infert <- read.csv("infertility.csv")
infert$education <- factor(infert$education)

head(infert)
```

```
      education age parity induced case spontaneous stratum
1 0-5 years   26     6     2     0           0           1
2 0-5 years   26     6     2     0           0           1
3 0-5 years   26     6     1     1           2           1
4 0-5 years   42     1     0     0           0           2
5 0-5 years   42     1     0     0           0           2
6 0-5 years   42     1     1     1           0           2
```

```
# One row per person
```

```
table(infert$case) # 83 cases, 165 controls
```

```
      0     1
165  83
```

```
length(unique(infert$stratum)) # 83 strata
```

```
[1] 83
```

```
table(table(infert$stratum)) # 1 of size 2, 82 of size 3
```

```
2 3
1 82
```

Question 6

We will examine the effect of spontaneous abortion on infertility using conditional logistic regression. Use `clogit` to perform a matched analysis using the variable `spontaneous` as a factor explanatory variable.

What do you conclude?

SOLUTIONS

```
cl_spon <- clogit( case ~ factor(spontaneous) + strata(stratum),
                  data = infert )
summary(cl_spon)
```

Call:

```
coxph(formula = Surv(rep(1, 248L), case) ~ factor(spontaneous) +
      strata(stratum), data = infert, method = "exact")
```

n= 248, number of events= 83

	coef	exp(coef)	se(coef)	z	Pr(> z)
factor(spontaneous)1	1.3288	3.7764	0.3673	3.617	0.000298 ***
factor(spontaneous)2	2.2740	9.7186	0.4785	4.753	2.01e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
factor(spontaneous)1	3.776	0.2648	1.838	7.758
factor(spontaneous)2	9.719	0.1029	3.805	24.825

Concordance= 0.685 (se = 0.046)

Likelihood ratio test= 34.05 on 2 df, p=4e-08

Wald test = 26.46 on 2 df, p=2e-06

Score (logrank) test = 34.27 on 2 df, p=4e-08

Having one previous spontaneous abortion multiplies the odds of infertility by 3.78 (95% CI: 1.84 to 7.76) conditional on age, parity and education. Women who have had two or more previous events have odds of infertility 9.72 (95% CI: 3.80 to 24.82) times that of women who have not had a spontaneous abortion conditioning on age, parity and education. A likelihood ratio test (the third line from the bottom in the model summary) provides very strong evidence ($p < 0.0001$) for the (conditional on age, parity and education) association between this categorical/factor spontaneous abortion variable and infertility.

Question 7

As discussed in the lecture, an alternative analysis for a matched case-control study with ‘well-defined’ matching variables is to perform an unmatched analysis with regression adjustment for the matching variables. Fit the following three models using `glm`.

- A model that only includes the matching variables (age, parity and education level),
- A model that only includes the exposure of interest, the count of spontaneous abortions, as a categorical/factor explanatory variable,
- A model that includes the exposure of interest and the matching variables.

Discuss: Compare the estimated coefficients for the matching variables in the models in parts (a) and (c). What do you conclude?

Also compare the estimated coefficients relating to the categorised count of spontaneous abortions in (b) and (c) with those from the matched analysis using conditional logistic regression. Which approaches give valid estimates of the odds ratios?

SOLUTIONS

```
### (a)
# Matching variables only
logit1 <- glm( case ~ age + parity + education,
              family = binomial,
              data = infert)
summary(logit1)
```

Call:

```
glm(formula = case ~ age + parity + education, family = binomial,
    data = infert)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.847542	1.245895	-0.680	0.496

age	0.002076	0.027245	0.076	0.939
parity	0.019070	0.117221	0.163	0.871
education12+ years	0.069988	0.718125	0.097	0.922
education6-11 years	0.046079	0.693754	0.066	0.947

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 316.17 on 247 degrees of freedom
 Residual deviance: 316.14 on 243 degrees of freedom
 AIC: 326.14

Number of Fisher Scoring iterations: 4

```
exp(coef(logit1))
```

	(Intercept)	age	parity	education12+ years
	0.4284667	1.0020779	1.0192527	1.0724948
education6-11 years	1.0471572			

Here all of the estimated odds ratios are close to 1 because these are all matching variables, so the variables are independent of case/control status.

```
### (b)
# Exposure of interest only
logit2 <- glm( case ~ factor(spontaneous),
              family = binomial,
              data = infert)
summary(logit2)
```

Call:

```
glm(formula = case ~ factor(spontaneous), family = binomial,
    data = infert)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.3952	0.2111	-6.609	3.87e-11 ***
factor(spontaneous)1	1.1403	0.3191	3.574	0.000352 ***
factor(spontaneous)2	2.0883	0.4118	5.071	3.95e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 316.17 on 247 degrees of freedom
Residual deviance: 283.67 on 245 degrees of freedom
AIC: 289.67

Number of Fisher Scoring iterations: 4

```
exp(coef(logit2))
```

(Intercept)	factor(spontaneous)1	factor(spontaneous)2
0.2477876	3.1276786	8.0714286

This is an invalid analysis because the matching has not been allowed for (either by adjustment, or by using conditional logistic regression).

```
### (c)
# Matched variables and exposure of interest
logit3 <- glm( case ~ factor(spontaneous) +
               age + parity +
               education,
               family = binomial,
               data = infer )
summary(logit3)
```

Call:

```
glm(formula = case ~ factor(spontaneous) + age + parity + education,
     family = binomial, data = infer)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.67566	1.35474	-0.499	0.6180
factor(spontaneous)1	1.31615	0.33822	3.891	9.97e-05 ***
factor(spontaneous)2	2.53274	0.47013	5.387	7.15e-08 ***
age	0.01628	0.02962	0.550	0.5826
parity	-0.30232	0.13834	-2.185	0.0289 *
education12+ years	-0.84628	0.79217	-1.068	0.2854
education6-11 years	-0.69467	0.75972	-0.914	0.3605

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 316.17 on 247 degrees of freedom
Residual deviance: 278.24 on 241 degrees of freedom
AIC: 292.24

Number of Fisher Scoring iterations: 4

```
exp(coef(logit3))
```

(Intercept)	factor(spontaneous)1	factor(spontaneous)2
0.5088224	3.7290433	12.5879383
age	parity	education12+ years
1.0164117	0.7390994	0.4290066
education6-11 years		
0.4992380		

The estimated odds ratios for the matching variables were close to 1 in a), but not in c). Unconditionally there is (little or) no association between the matching variables and case-control status by design, but once we include the exposure (here the spontaneous abortion variable) in the model there are (study design induced) associations that must be allowed for. This illustrates that we must always adjust for matching variables either explicitly (as here), or implicitly using conditional logistic regression.

The analysis in c) is valid, whereas (as stated above) that in b) is invalid because the matching has not been allowed for (either by adjustment, or by using conditional logistic regression). For the valid approach in c) the standard errors on the spontaneous abortion parameter estimates are slightly narrower than with the conditional logistic regression model (`cl_spon`), possibly reflecting improved statistical efficiency. However, the conditional logistic regression model avoids having to make choices about the nature of the relationship between matching variables and the outcome (here they are assumed linear on a logit scale).